SCIENCE OLYMPIAD FOUNDATION

Class 8

IMO

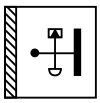
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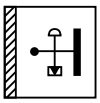
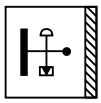
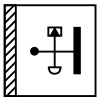
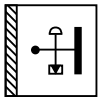
Year 2018

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1. In a certain code language, if 'CONFIDENCE' is coded as 'FGPEGEQPHK', then how will 'PERFECTION' be coded in the same language?
   A. RGTHGEVKQP
   B. EVKQPRGTHG
   C. DUJPOQFSGF
   D. PEFRECITON

2. Select the correct water image of the given figure.



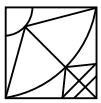
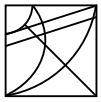
   A. 
   B. 
   C. 
   D. 

3. Starting from a point P, Aryan walked 40 metres towards North. He turned right and walked 60 metres. He then turned right and walked 40 metres. At last he turned right and walked 40 metres and reached a point Q. How far and in which direction is he now from the starting point?
   A. 10 metres, West
   B. 20 metres, West
   C. 20 metres, East
   D. 10 metres, East

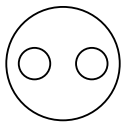
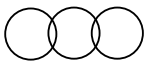
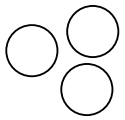
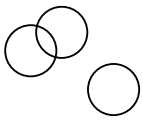
4. If it is possible to make only one meaningful word with 2nd, 4th, 6th and 11th letters of the word 'INTELLIGENT' (using each letter only once), then third letter of the word formed would be your answer. If more than one such words can be formed, then give Z as the answer and if no such word can be formed, then give Y as the answer.
   A. N
   B. Y
   C. Z
   D. E

5. Select a figure from the options in which the given figure is exactly embedded as one of its parts.

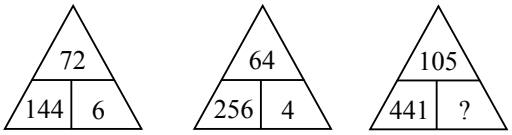


   A. 
   B. 
   C. 
   D. 

6. Select the odd one out.
   A. Z 25 X 23
   B. P 15 N 13
   C. H 7 F 4
   D. D 3 B 1

7. Which of the following Venn diagrams best represents the relationship amongst, 'Shirts, Trousers and Skirts'?
   A. 
   B. 
   C. 
   D. 

8. Find the missing number, if a same rule is followed in all the three figures.

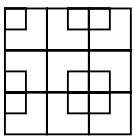


   A. 11
   B. 7
   C. 5
   D. 9

9. A family consists of five members B, C, E, F and G. C is the son of G but G is not the mother of C. E and G are married couple. B is the brother of G. F is the daughter of E. Who is the brother-in-law of E?
   A. F
   B. G
   C. B
   D. None of these

10. If '–' denotes '+', '×' denotes '÷', '+' denotes '×', '=' denotes '<' and '÷' denotes '–', then which of the following options is correct?
    A. $6 + 4 \div 20 - 12 \times 6 = 2$
    B. $3 \div 3 \times 3 - 4 + 5 = 20$
    C. $12 \times 6 + 4 \div 3 - 6 = 10$
    D. $4 - 20 \times 5 + 4 \div 4 = 20$

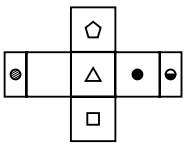
11. How many squares are there in the given figure?



    A. 24
    B. 26
    C. 20
    D. None of these

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12. Choose a box that is similar to the box formed when the given sheet of paper is folded to form a box.



(I) 

(II) 

(III) 

(IV) 

A. I only
B. I and III only
C. IV only
D. II and IV only

13. Two rows of numbers are given. The resultant number in each row is to be worked out separately based on the following rules and the question below the rows of numbers is to be answered. The operations on numbers progress from left to right.

Rules:

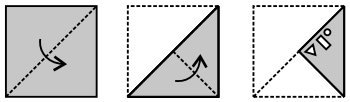
(i) If a 2-digit even number is followed by another even number, then the first one is to be divided by the second one.
(ii) If an even number is followed by a prime number, then the two are to be multiplied.
(iii) If an odd number is followed by another odd number, then the two are to be added.
(iv) If a 3-digit number is followed by a 2-digit number which is a perfect square, then the second number is to be subtracted from the first number.
(v) If a 3-digit number is followed by another 3-digit number which is not a perfect square, then the first number is to be divided by the second one.

150	81	43
336	*m*	39

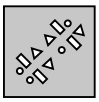
If *m* is the resultant of the first row, then what will be the resultant of the second row?

A. 42
B. 112
C. 40
D. 36

14. The question consists of a set of three figures P, Q and R showing a sequence of folding a piece of paper. Fig. (R) shows the manner in which the folded paper has been cut. Select the figure from the options which would most closely resembles the unfolded form of Fig. (R).



P Q R

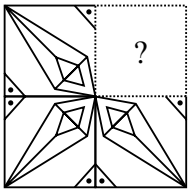
A. 

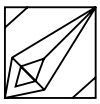
B. 

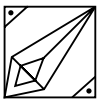
C. 

D. 

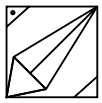
15. Select a figure from the options which will replace the '?' to complete the pattern in the given figure.



A. 

B. 

C. 

D. 

MATHEMATICAL REASONING

16. Which of the following is equal to 25?

A. $\dfrac{(0.22)^2}{0.2\times(2.2)^2}$

B. $\dfrac{(0.022)^2}{2.2\times0.022}$

C. $\dfrac{(2.2)^2}{22^2\times(0.02)^2}$

D. $\dfrac{(0.22)^2}{22^2\times0.02}$

17. If $\left(\dfrac{1}{16}\right)^{4-3x}\times8^{x-2}=(0.25)^x$, then find the value of $\dfrac{17x}{22}+1$.

A. 2
B. $\dfrac{39}{22}$
C. 1
D. $\dfrac{22}{17}$

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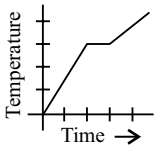
18. Given that the number 67y19 is divisible by 9, where y is a single digit, what is the least possible value of y?

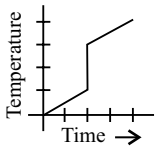
A. 3
B. 9
C. 7
D. 4

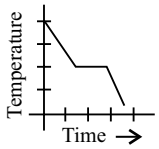
19. Which of the following rational numbers does not lie between $\frac{1}{4}$ and $\frac{2}{3}$?

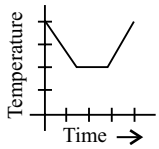
A. $\frac{1}{2}$
B. $\frac{1}{3}$
C. $\frac{14}{24}$
D. $\frac{18}{24}$

20. Which of the following graphs cannot be a time temperature graph?

A. 

B. 

C. 

D. 

21. How many cubes of volume 216 cm^3 can be obtained from a cube whose side is 36 cm ?

A. 216
B. 125
C. 729
D. 343

22. The numerator and the denominator of a rational number are in the ratio 5 : 7. When 6 is added to both the numerator and denominator, the ratio becomes 4 : 5. What is the rational number?

A. $\frac{7}{5}$
B. $\frac{5}{7}$
C. $\frac{2}{5}$
D. $\frac{13}{14}$

23. If $x^2+\frac{1}{x^2}=\frac{17}{4}$, then find the value of $\frac{2}{5}\left(x+\frac{1}{x}\right)+\left(x-\frac{1}{x}\right)$.

A. $\frac{3}{2}$
B. $\frac{25}{4}$
C. $\frac{5}{2}$
D. $\frac{9}{4}$

24. Simple interest on a sum of money for 1 year at 12% per annum is ₹ 1200. What will be the compound interest when compounded half-yearly on that sum at the same rate for the same period?

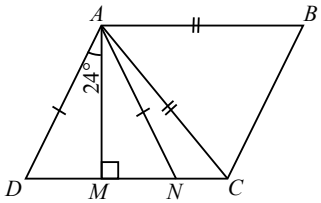
A. ₹ 1236
B. ₹ 1326
C. ₹ 11236
D. ₹ 13260

25. For what value of P, the given equation is correct?

$$\frac{1-P}{6}+\frac{2P}{3}-\frac{1-7P}{4}=2\frac{1}{6}$$

A. –1
B. 2
C. –2
D. 1

26. If $ABCD$ is a parallelogram, ΔADN and ΔABC are isosceles triangles, then find $\angle BAC$.



A. 112°
B. 140°
C. 48°
D. 32°

27. Given below are the steps of construction to construct a quadrilateral $KLMN$ where KL = 6 cm, LM = 8 cm, MN = 9.5 cm, $\angle L$ = 105° and $\angle M$ = 75°. Which of the following steps is incorrect?

Step-1 : Draw LM = 8 cm and construct $\angle MLX$ = 105°.
Step-2 : With L as centre and radius 6 cm, draw an arc which cuts LX at K.
Step-3 : Construct $\angle LMY$ = 75°. With M as centre and radius 9.5 cm, draw an arc which cuts MY at N.
Step-4 : Now join N and X to obtain the required quadrilateral $KLMN$.

A. Step-3
B. Step-1
C. Step-2
D. Step-4

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28. If a and b vary inversely to each other, then find the value of $\frac{p+q}{2}$.

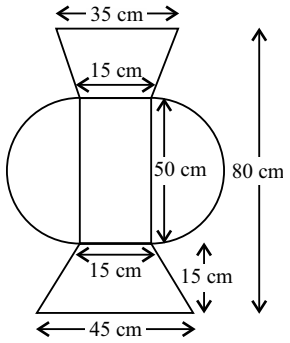
a	6	p	90
b	q	5	2

A. 30
B. 33
C. 36
D. 66

29. Divide $(y^2 + 6y - 16)(y + 2)$ by $(y^2 - 4)$.
A. $y + 8$
B. $y - 2$
C. $y + 4$
D. $2y + 4$

30. When a dice is thrown, find the probability of not getting a prime number.
A. $\frac{1}{6}$
B. $\frac{3}{5}$
C. $\frac{1}{2}$
D. $\frac{1}{5}$

31. Find the area of the given figure (not drawn to scale).

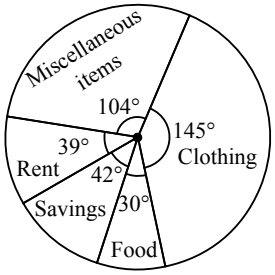


A. 3339.29 cm^2
B. 3539.29 cm^2
C. 4506.75 cm^2
D. 5967.47 cm^2

32. If a four-digit number $ccdd$ is divisible by 55, then the possible value(s) of d is/are_____.
A. Either 0 or 2
B. Either 5 or 2
C. 5
D. Either 0 or 5

33. Select the incorrect match of given solids with the product of their number of faces and vertices.
A. Rectangular pyramid – 25
B. Triangular prism – 30
C. Octahedron – 48
D. Triangular pyramid – 18

Direction (34-35): The given pie chart shows the monthly expenditure of a family on rent, clothing, food, savings and miscellaneous items. The family spends ₹ 520 on miscellaneous items.



34. How much more does the family spend on clothing than on savings?
A. ₹ 752
B. ₹ 515
C. ₹ 650
D. ₹ 725

35. How much does the family spend on rent, food and clothing in a year, if the expenditure is same throughout the year?
A. ₹ 12840
B. ₹ 14820
C. ₹ 15760
D. ₹ 18420

EVERYDAY MATHEMATICS

36. A water pump pumps out $14\frac{1}{6}$ L of water per minute from a reservoir. How many litres of water will be pumped out in $1\frac{1}{5}$ of an hour?
A. 1125 L
B. 6120 L
C. 1020 L
D. 1560 L

37. How many bricks of size 22 cm × 10 cm × 7 cm are required to construct a wall 11 m long, 3.5 m high and 40 cm thick, if the cement and sand used in the construction occupy $(1/10)^{th}$ part of the wall?
A. 8000
B. 9000
C. 7000
D. 10000

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38. A man borrowed ₹ 4000 at 10% per annum compound interest. At the end of each year he has repaid ₹ 1000. The amount of money he still incurs after the third year is _____.

A. ₹ 2740
B. ₹ 2104
C. ₹ 2014
D. ₹ 3400

39. Vikas's garden is in the form of a parallelogram whose one side is 4.8 cm and other side is $1\frac{1}{2}$ times of this side. He wants to fence his garden four times by a wire. Find the length of the wire required.

A. 96 cm
B. 108 cm
C. 132 cm
D. 84 cm

40. Ritesh selected a card from a deck of 52 playing cards. Find the probability that the selected card is a red queen.

A. $\frac{2}{13}$
B. $\frac{1}{24}$
C. $\frac{1}{13}$
D. $\frac{1}{26}$

41. Each member of a party contributed thrice as many rupees as the total number of members and the total collection was ₹ 7203. How many members contributed for the party?

A. 49
B. 39
C. 26
D. 46

42. If 20 pumps of equal capacity can fill a tank in 7 days, then how many more such pumps will be required to fill the tank in 5 days?

A. 28
B. 16
C. 8
D. 7

43. Two years ago, Mohit was three times as old as his son and two years hence, twice of Mohit's age will be equal to five times that of his son. The present age of Mohit is _____.

A. 14 years
B. 38 years
C. 32 years
D. 34 years

44. The circumference of the base of a circular cylinder is 8 π cm. The height of the cylinder is same as the diameter of the base. How much water can the cylinder hold?

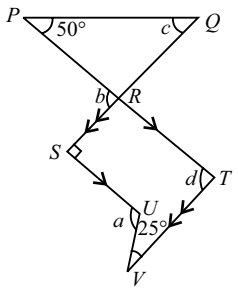
A. 128 π litres
B. 0.128 π litres
C. 630 π litres
D. 0.164 π litres

45. A shopkeeper fixes the marked price of a pair of shoes 45% above its cost price. What is the percentage of discount allowed to gain 16%?

A. 30%
B. 25%
C. 15%
D. 20%

ACHIEVERS SECTION

46. In the given figure (not drawn to scale), find the value of $(b + d) - (a + c)$.



A. 115°
B. 40°
C. 90°
D. 25°

47. Read the statements carefully and select the correct option.

Statement-1: The amount of extension in an elastic spring varies directly as the weight hung on it. If a weight of 150 gm produces an extension of 2.9 cm, then the weight that would produce an extension of 17.4 cm is 900 gm.

Statement-2: x and y vary directly means if x increases then y decreases.

A. Both Statement-1 and Statement-2 are true.
B. Both Statement-1 and Statement-2 are false.
C. Statement-1 is false but Statement-2 is true.
D. Statement-1 is true but Statement-2 is false.

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48. A box has 30 balls with numbers 1 to 30 written on them. One ball is drawn at random. Find the value of $(P + Q) \div R$ by completing the given table.

	Event	Probability
(i)	Getting a ball with a number less than 31	P
(ii)	Getting a ball with a number greater than 30	Q
(iii)	Getting a ball with an odd number	R

A. 2

B. $\frac{1}{2}$

C. 1

D. 0

49. Fill in the blanks.

(i) If each edge of a cube is doubled, then the surface area of the new cube will become __**P**__ times.

(ii) Multiplicative inverse of 2^{-16} is __**Q**__.

(iii) The unit digit of the cube of the number 1787 is __**R**__.

(iv) If three acute angles of a quadrilateral measure 70° each, then the measure of the fourth angle is __**S**__.

	P	**Q**	**R**	**S**
A.	Two	2^{-16}	3	140°
B.	Four	2^{16}	3	150°
C.	Three	$\frac{1}{2^{16}}$	9	110°
D.	Two	$\frac{1}{2^{-16}}$	9	150°

50. State 'T' for true and 'F' for false and select the correct option.

(i) The measure of each exterior angle of an n-sided regular polygon is $\left(\frac{180^\circ}{n}\right)$.

(ii) The adjacent angles in a parallelogram are supplementary.

(iii) In a quadrilateral $PQRS$, the bisector of $\angle P$ and $\angle Q$ meet at point M. If $\angle R = 120°$ and $\angle S = 40°$, then $\angle PMQ = 80°$.

	(i)	**(ii)**	**(iii)**
A.	F	F	T
B.	F	T	T
C.	T	F	F
D.	T	T	F

SPACE FOR ROUGH WORK

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